

Question 1

- 1.1. A. **Improved experience:** CloudFront caches video content at edge locations close to users in Africa, Europe, and the Middle East. This reduces round-trip latency and minimizes buffering.
- b. **Reduced latency** Instead of every request going back to the origin S3 bucket, viewers fetch content from the nearest edge location. This shortens delivery paths and improves streaming quality during live events.
- c. **Scalability** Edge caching absorbs traffic spikes, preventing overload on the origin servers.
- d. **TCP Optimization** Cloudfront optimizes TCP connections to reduce the handshake time for each user request, accelerating the start of video playback.
- e. **Durable Streaming** For lives event, Cloudfront supports features like serving live video with automatic bitrate switching, ensuring that users with varying internet speeds can still watch without buffering by using the appropriate bitrate from the edge.

1.2. Challenge 1:

1. Bandwidth costs during live events

- Thousands of simultaneous viewers increase CloudFront data transfer costs.
- Recommendation:

Enable CloudFront caching policies and Origin Shield to reduce redundant requests. Use Reserved Capacity pricing for predictable traffic.

2. Cache inefficiency(performance)

- During major live events, thousands of users might be requesting the same content, but if the cache hit ratio low, lambda and s3 will be overwhelmed, causing high latency and high origin fetch costs.
- Recommendations:

Implement cloudfront origin shield, this adds an additional caching layer between the edge locations and the s3 origin. It consolidates request from multiple edge locations, significantly increasing the cache hit ratio at the origin, reducing load on s3, and lowering the number of requests to the bucket.

Challenge 2:

1.DynamoDB scaling costs

- Rapid growth in user sessions and preferences can lead to high read/write costs.
- Recommendation:

Use DynamoDB Auto Scaling with On-Demand capacity mode to handle unpredictable spikes efficiently. Implement DynamoDB Accelerator for faster reads.

Question 2

Given:

The following pricing assumptions can be considered:

- Total hours: 730 hours/month
- EC2 t3.large: ~\$0.0832/hour
- EBS gp3: ~\$0.08/GB/month
- S3 Standard: ~\$0.023/GB/month
- EC2 outbound transfer: ~\$0.09/GB after first 1 GB
- CloudFront transfer: ~\$0.085/GB
- AWS Systems Manager Advanced Tier: ~\$0.00695 per managed instance/h

2.1.

Question 3

3.1. VPC

The screenshot shows the 'Create VPC' page in the AWS Management Console. The left sidebar contains the 'VPC settings' section with the following options:

- Resources to create:** ☐ VPC only, ☒ VPC and more
- Name tag auto-generation:** ☒ Auto-generate, ShopEase-VPC
- IPv4 CIDR block:** 10.10.0.0/16 (65,536 IPs)
- IPv6 CIDR block:** ☒ No IPv6 CIDR block
- Tenancy:** Default
- Encryption settings - optional:** (collapsed)
- Number of Availability Zones (AZs):** (collapsed)

The right sidebar shows a 'Preview' section with a diagram of the VPC architecture:

- VPC:** ShopEase-VPC-vpc
- Subnets (4):**
 - us-east-1a: ShopEase-VPC-subnet-public1-us-east-1a, ShopEase-VPC-subnet-private1-us-east-1a
 - us-east-1b: ShopEase-VPC-subnet-public2-us-east-1b, ShopEase-VPC-subnet-private2-us-east-1b
- Route tables (3):** ShopEase-VPC-rtb-public, ShopEase-VPC-rtb-private1-us-east-1a, ShopEase-VPC-rtb-private2-us-east-1b

3.2. Public Subnets and Private Subnets

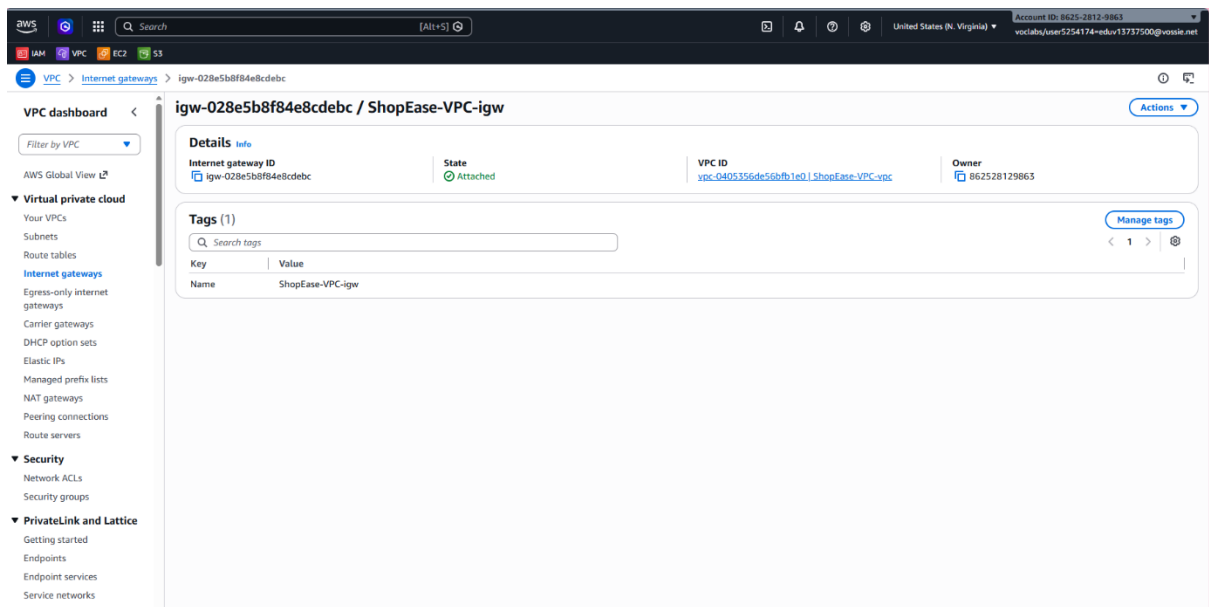
The screenshot shows the 'Create VPC' page with the 'Customize AZs' section expanded. The left sidebar contains the following options:

- Number of public subnets:** 0, 2 (selected)
- Number of private subnets:** 0, 2 (selected), 4
- Customize subnets CIDR blocks:**
 - Public subnet CIDR block in us-east-1a: 10.10.1.0/24 (256 IPs)
 - Public subnet CIDR block in us-east-1b: 10.10.2.0/24 (256 IPs)
 - Private subnet CIDR block in us-east-1a: 10.10.10.0/24 (256 IPs)
 - Private subnet CIDR block in us-east-1b: 10.10.20.0/24 (256 IPs)
- NAT gateways (\$):** ☒ None, ☐ Regional - new, ☐ Zonal
- Introducing regional NAT gateway:** (button)

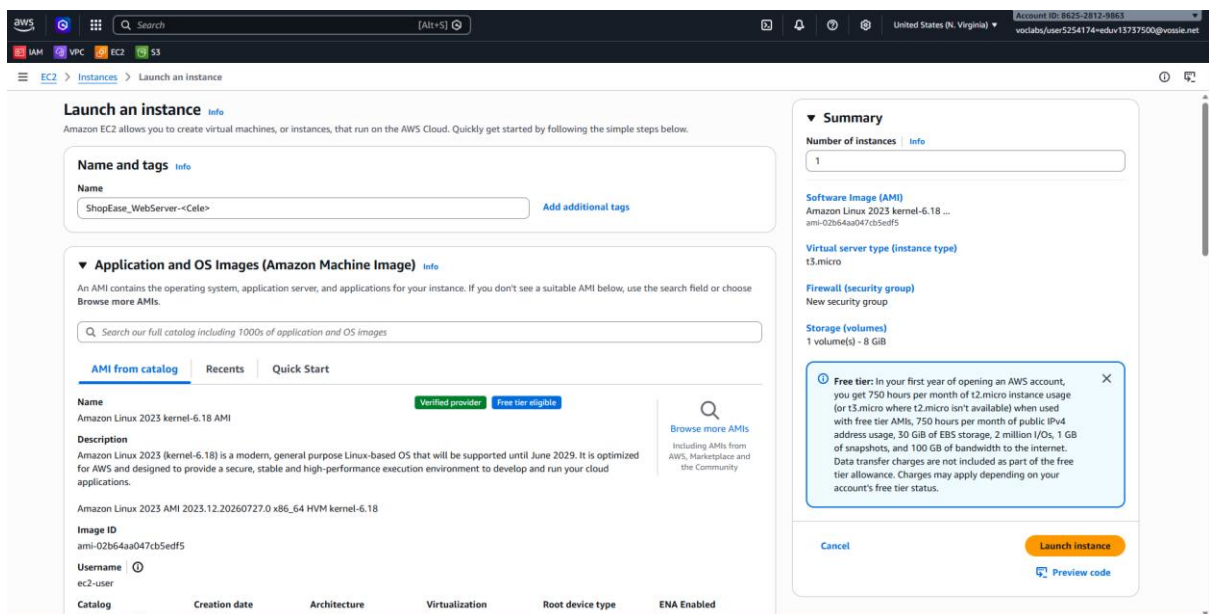
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- Route tables (3):** ShopEase-VPC-rtb-public, ShopEase-VPC-rtb-private1-us-east-1a, ShopEase-VPC-rtb-private2-us-east-1b

3.3. internet Gateway



3.4.



3.5.

3.6. These components from layered security model for the ShopEase infrastructure. Public and Private subnets, combined with associated route tables, control where traffic can reach the Public-RT routes traffic from the public subnets to the internet via the internet gateway, enabling the web servers to be reached externally, while the private-RT deliberately excludes an internet routes, ensuring the private application subnets remain isolated from direct external access. This network -level segmentation reflects the principle of separating public- facing and internal resources.

security groups then acts as a stateful firewall at the instance level, controlling what traffic is permitted in this case, SG WEB only allows inbound SSH, HTTP, and HTTPS traffic, blocking all other unsolicited traffic by default.

IAM governs who is permitted to access and modify these AWS resources in the first place, ensuring that only authorised administrators can create, alter or delete infrastructure components.